

1. To make "python main.py" run and returns results – I needed to change the ports on main.py file from:

```
# request_current_from_ammeter(5001, b'MEASURE_GREENLEE') #  
Request from Greenlee Ammeter  
# request_current_from_ammeter(5002, b'MEASURE_ENTES') #  
Request from ENTES Ammeter  
# request_current_from_ammeter(5003, b'MEASURE_CIRCUTOR') #  
Request from CIRCUTOR Ammeter
```

To:

```
request_current_from_ammeter(5000, b'MEASURE_GREENLEE -  
get_measurement') # Request from Greenlee Ammeter  
request_current_from_ammeter(5001, b'MEASURE_ENTES -  
get_data') # Request from ENTES Ammeter  
request_current_from_ammeter(5002, b'MEASURE_CIRCUTOR -  
get_measurement') # Request from CIRCUTOR Ammeter
```

According to the readme file, I changed both ports and commands and then saw the results from the ammeters
also removed the comment from this code:

```
greenlee:  
  port: 5000  
  command: "MEASURE_GREENLEE -get_measurement"  
entes:  
  port: 5001  
  command: "MEASURE_ENTES -get_data"  
circutor:  
  port: 5002  
  command: "MEASURE_CIRCUTOR -get_measurement"
```

2. Removed the -current flag from the circutor ammeter file, the logic of the measure didn't change. Now its just like the readme defined it
3. Changed main.py ports to fit the other files ports:

```
def run_greenlee_emulator():  
    greenlee = GreenleeAmmeter(5000)  
    greenlee.start_server()  
  
def run_entes_emulator():  
    entes = EntesAmmeter(5001)  
    entes.start_server()  
  
def run_circutor_emulator():
```

```
circutor = CircutorAmmeter(5002)
circutor.start_server()
```

4. At this point, I did not need to add any new libraries
5. Unified Measurement API – changed request_current_from_ammeter function to return the data instead only print it. That's way we can get the data for analyzing the results in our tests. Added test_framework.py file – we load the config file, get the ammeters from there (and the ports and commands), do the request using request_current_from_ammeter and return the response.
6. Measurement Sampling – changed the test_framework.py file to not only call once to request_current_from_ammeter but using this 3 params from the config file + changed the config file to have default parameters.

```
testing:
    sampling:
        measurements_count: 5
        total_duration_seconds: 4
        sampling_frequency_hz: 1
```

Added import time to test_framework.py file

7. Result Analysis – added import to statistics library for calculating mean, median, and standard deviation, also the _analyze_measurements function returns the min and max values.

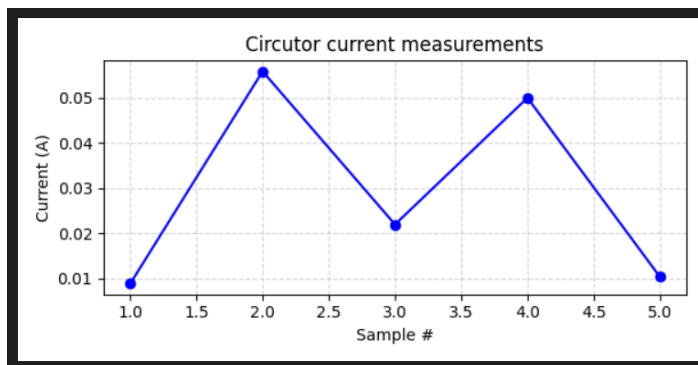
```
return {
    "count": len(values),
    "mean": statistics.mean(values),
    "median": statistics.median(values),
    "standard_deviation": statistics.stdev(values) if
len(values) > 1 else 0.0,
    "minimum": min(values),
    "maximum": max(values),
}
```

8. Imported pathlib for get to the root dir of the project and create there report dir, the main goal is to save the results there.
9. python -m pip install pyyaml – handle config file
python -m pip install matplotlib – for png plots
10. needed the servers to stay alive so I can run the tests file, so I added while loop to run until pause, I also changed the threads to be regular threads so they keep the socket for all ports open all the time.

11. I updated reame file with the correct commands for the tests + the graph and the results.
12. python examples/run_tests.py -> results on terminal + pictures at report directory (currently override between runs).

Results for example (we have 1 for each type):

```
Results for circutor:  
port: 5002  
command: MEASURE_CIRCUTOR -get_measurement  
status: ok  
final measurement: 0.010334408979728787 A  
raw response: 0.010334408979728787  
analysis:  
  count: 5  
  mean: 0.029366334058891996  
  median: 0.02192607999459325  
  standard_deviation: 0.022162319943663866  
  minimum: 0.008793147496837822  
  maximum: 0.055822694286645935  
consistency:  
  stable: False  
  range: 0.04702954678980811  
  coefficient_of_variation: 0.7546845956059406  
  trend: increasing  
  remarks: unstable  
plot saved to: C:\Users\aviga\Downloads\PyTest_QA_u  
reports\circutor_measurements.png
```



13. Result Management – when we run "python examples/run_tests.py" (after python main.py) we can see in results directory new json file that contains the data of the all ammeters that tested. I decided to put them together in the same file, but we could of create 3 different files/ unique directory with 3 files in it.